



Generative Ai



with step

- 1
- 2 cup maida
- 3 table spoon sugar



Prompt

Pneumonia

Task

Context

You are a witty social media Expert. Give me a funny Instagram caption about happily coding for 12 hours straight.



→ Advance Prompt Engineering

Persona

category 1

Expertise
level



category 2

Role based
Persona



category 3

Personality
+
tone



category 4

Biased &
ethical
constraints



1

Expertise Level



2

Role based Persona



3

Personality and tone

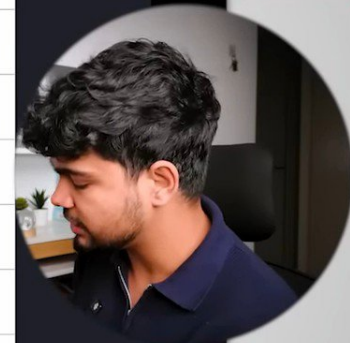


4

Biases and ethical constraints



Task Subcategories			
Response Type Summarization → Summarize this article in 3 bullet points. Explanation → Explain in simple terms like "I'm 5 years old." Creative Writing → Write a sci-fi short story in the style of Isaac Asimov." Comparison → Compare the iPhone 15 and Samsung S24 based on features.	Output Formatting Define the structure of AI's response. Examples → Give me a response in a JSON format. → Write this answer as a bulleted list. → Generate a table with key differences between Python and Java.	Depth & Length Control Specify how detailed the response should be. Examples → Answer in one sentence. → Give a detailed answer with at least 5 supporting examples. → Explain in a tweet-sized response	Creativity vs. Precision Adjust how rigid or creative the AI should be. Examples → Be highly factual and cite sources. → Be creative and add a metaphor.



Context Subcategories				
Domain Context Defines the industry or field AI should focus on. <u>Examples</u> Explain AI in the context of healthcare. → Generate marketing strategies for an e-commerce startup." → Talk about cybersecurity risks in cloud computing.	Prior Information Carry over previous instructions or context. <u>Examples</u> → Using the last example you gave, now apply it to finance. → Remember the company I mentioned earlier? Generate a marketing strategy for it. → Based on the email draft you created, now refine it for a CEO audience.	Target Audience Tailor responses for specific user groups. <u>Examples</u> → Explain this topic for 8th-grade students. → Write an article on AI trends for a business magazine. → Summarize this research paper for non-technical professionals.	Cultural & Regional Adaptation Ensure AI localizes responses for different audiences. <u>Examples</u> → Explain in Hinglish with Indian cultural references. → Use examples relevant to European startups. → Make it relatable for an American college student.	Style & Mood Context Fine-tune writing style or emotional tone. <u>Examples</u> → Make it sound like a TED Talk speech. → Write in an emotional and inspiring tone. → Keep it neutral and purely analytical.



1

Chain of thoughts Prompting



Meaning of Life Exploration

Gemini

gemini.google.com/app/d263341b59aaf028

Finish update

Gemini

2.0 Flash

Try Gemini Advanced

what is 487 x 27

487 x 27 = 13149

Ask Gemini

Gemini can make mistakes, so double-check it



Meaning of Life Exploration

Gemini

gemini.google.com/app/d263341b59aaf028

Finish update

Gemini 2.0 Flash

Try Gemini Advanced

Think step by step. First, break down 487×27 into two smaller multiplications

Just a second...

+

Ask Gemini

Gemini can make mistakes, so double-check it



2

Few Shot Prompting



Basics of Prompt Engineering

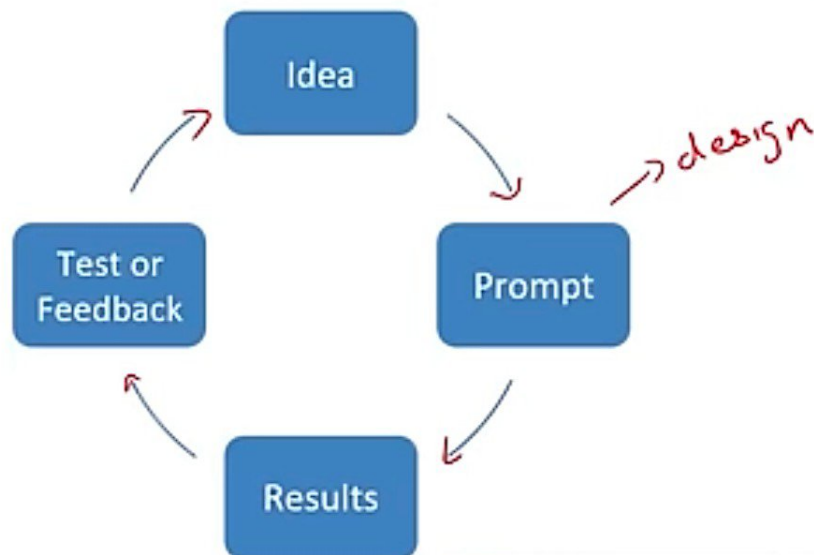
{ Prompt engineering is as much an art as it is a science. }

Prompt: A detailed set of guidelines given to an LLM to do a task.

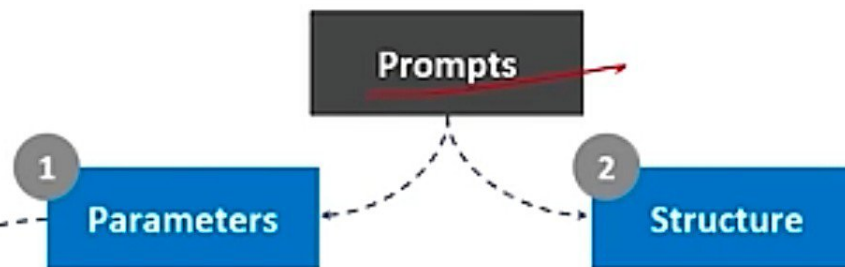
Engineering: Developing a task-specific prompt iteratively.

Prompt Engineering is an iterative process.

Code generation



Basics of Prompt Engineering



1. Temperature: Lower for creativity and deterministic results; Higher for diversity and factual QA.
2. Top P: Control model determinism; Low for factual, high for diverse.
3. Max Length: Manage response length.



Components of a Good Prompt

Context

Additional context that can steer the model to better responses.

Instruction

A specific task or instruction you want the model to perform.

Input Data

The input or question that we are interested in finding a response for.

Output Indicator

The type or format of the output.



Components of a Good Prompt

Context

Additional context that can steer the model to better responses.

Context

Act as an analyst working for an OTT platform. You will have to perform sentiment analysis based on the feedback provided by the consumers on the movies and series put on the OTT platform.



Components of a Good Prompt

Context

Additional context that can steer the model to better responses.

Instruction

A specific task or instruction you want the model to perform.

Instruction

Classify the feedback as neutral, negative, or positive, where positive is promotor; negative means demoter, and neutral means neither will promote nor demote the content.



Components of a Good Prompt

Context

Additional context that can steer the model to better responses.

Instruction

A specific task or instruction you want the model to perform.

Input Data

The input or question that we are interested in finding a response for.

Output Indicator

The type or format of the output.

Example

Feedback: I think the series was okay.

Sentiment: Neutral

Feedback: The acting of each character in the series was awesome.

Sentiment: Positive

Input Data and Output Indicator

Feedback: The storyline for the series was repetitive and abysmal.

Sentiment:



Components of a Good Prompt

Context

Act as an analyst working for an OTT platform. You will have to perform sentiment analysis based on the feedback provided by the consumers on the movies and series put on the OTT platform.

Instruction

Classify the feedback as neutral, negative, or positive, where positive is promotor; negative means demoter, and neutral means neither will promote nor demote the content.

Example

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Sentiment: Neutral

Feedback: The acting of each character in the series was awesome.

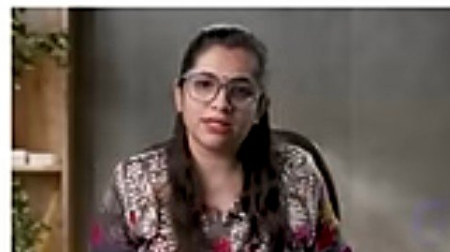
Sentiment: Positive

Input Data and Output Indicator

Feedback: The storyline for the series was repetitive and abysmal.

Sentiment:

Negative



How to Write a Good Prompt?

Here's a checklist, to create the most effective prompt and get the best results.

- **Define the Goal** – Tell ChatGPT what you exactly want it to do.
- **Detail out the Format** – Specify the format in which you want your output. E.g., tables/paragraphs/lists, with or without a heading, listed in priority order if any, etc.
- **Create a Role** – Assign ChatGPT a role to let it process your request from that specific point of view. E.g., Act as X.
- **Clarify who the Audience is** – Specify the demographics for ChatGPT to help it tailor its response appropriately.
- **Give Context** – Provide every possible information to ChatGPT to help it understand the purpose of your request.



How to Write a Good Prompt?

Here's a checklist, to create the most effective prompt and get the best results.

- **Give Examples** – Share examples to let ChatGPT learn from it and produce more accurate results.
- **Specify the Style** – Outline the tone, the communication style, the brand identity and other details in your prompt for a suitable response.
- **Define the Scope** – Outlining a scope with further specifications besides giving a context and examples, will help ChatGPT operate within those parameters.
- **Apply Restrictions** – Constraints or restrictions applied in your prompt will create the right boundaries for ChatGPT to produce more relevant responses.



Different Prompt Patterns

What are some of the most common Prompt Patterns?

1. **Persona Pattern** - Act as X. Do the task Y. (E.g., As a Yoga instructor, create a beginner-friendly routine for joint mobility.)
2. **Audience Persona Pattern** - Explain X to me. Assume I'm Y. (E.g., Explain the importance of eating greens. Assume I'm a skeptical child.)
3. **Visualization Generator Pattern** - Generate X that can be provided to tool Y for visualization. (E.g., While talking about the data around user engagement metrics on certain website, generate a CSV that I can use in Tableau to create a visualization.)



Different Prompt Patterns

What are some of the most common Prompt Patterns?

4. **Recipe Pattern** - In order to do X, I need to perform steps a, b, c. Provide a complete sequence of steps for me while filling in any missing steps and removing redundant steps. (Eg: I want to travel from Bangalore to Darjeeling. I know I have to take a flight to Kolkata. And from there take train and a cab to Darjeeling. Complete the itinerary for me).

5. **Template Pattern** - I will provide a template, with placeholders for content. Fit the output with one or more placeholders that I list.

Eg: Generate a day-wise travel itinerary for visiting Paris.

My placeholders are -

- ☐ <Day> for the day of the travel plan
- ☐ <Location> for the place to visit
- ☐ <Activity> for what to do in that place
- ☐ <Time> for the best part of the day to visit

Template: For <Day> visit <Location> at <Time> for <Activity>



Common Prompting Errors

1. Vague or Ambiguous Prompts
2. Biased Prompts
3. Lack of Contextual Information
4. Insufficient Examples or Training Data
5. Complex or Confusing Prompts
6. Not testing prompts thoroughly



Applications of Prompt Engineering

Content Generation

- Copywriting and Advertising ✓
- Creative Writing ✓
- Educational Content ✓

Customer Support and Engagement

- Chatbots and Virtual Assistants
- Instructional Guides

Data Analysis and Science

- Data Cleaning and Preparation
- Statistical Analysis and Visualization

Code Generation and Software Development

- Automating Code Writing
- Debugging Assistance
- Documentation Generation

Research and Information Retrieval

- Text Summarization
- Question Answering



Applications of Prompt Engineering

Machine Translation

- Information Sharing
- Internal Communication

Sentiment Analysis

- Consumer Feedback Analysis
- Brand Management
- Marketing

Other Domains

- Healthcare ✓
- Manufacturing
- Security
- Retail and Shopping



Advanced Prompt Strategies

Here's a few instances of advanced prompt strategies:

- **Zero-Shot Learning** - Directly instructing the model to perform a task without any additional examples.
- **Few-Shot Learning** – Teaching with examples in the prompt.
- **Chain of Thought Process** – Asking to show its work or thought process.

